Project Summary: House Price Prediction Using Machine Learning

This project focused on developing a machine learning model to predict house prices using various housing-related features. The dataset used for this study contained information about residential properties, including area, number of bedrooms, bathrooms, stories, parking spaces, availability of amenities such as air conditioning, guest rooms, basement, hot water heating, preferred area status, and furnishing status. The main objective was to analyze the factors affecting house prices and build predictive models that could estimate the selling price of a house based on these characteristics.

The project began with data loading and exploration using the Pandas library. The dataset was examined to understand its structure, identify the target variable (price), and detect any missing values or duplicate records. Data cleaning techniques were applied to handle missing values, remove duplicates, and convert categorical variables into numerical form using one-hot encoding. Outlier detection and removal were also performed to improve model performance and reduce the impact of extreme values on predictions.

Two machine learning algorithms were implemented for price prediction: **Linear Regression** and **Random Forest Regressor**. The dataset was divided into training and testing sets using an 80:20 ratio. The Linear Regression model was trained to establish a linear relationship between housing features and house prices, while the Random Forest model was used to capture more complex relationships through an ensemble of decision trees. The models were evaluated using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score.

The results showed that the **Linear Regression model outperformed the Random Forest Regressor** on this dataset. Linear Regression achieved an MAE of approximately **731,740**, an RMSE of approximately **971,860**, and an **R^2 score of 0.718**, indicating that the model explained about **71.8% of the variation in house prices**. The Random Forest model achieved an MAE of approximately **751,320**, an RMSE of approximately **1,024,107**, and an **R^2 score of 0.687**. These results indicate that the Linear Regression model provided more accurate predictions for this particular dataset.

Several visualizations were created to better understand the data and model performance. A histogram of house prices revealed the overall distribution of property values. A correlation heatmap helped identify the relationships between features and house prices. Scatter plots such as Price vs Area and Actual vs Predicted Price provided insights into the influence of housing characteristics and the effectiveness of the prediction models.

The analysis revealed that **area, bathrooms, parking spaces, air conditioning, and number of stories** were among the most important factors affecting house prices. Among all features, the **area of the house** had the strongest influence on the selling price. It was also observed that amenities such as air conditioning and preferred locations significantly increased property value. An interesting finding was that house area had a greater impact on price than the number of bedrooms alone.

In conclusion, this project successfully demonstrated the application of machine learning techniques for house price prediction. The Linear Regression model achieved good predictive performance and proved to be the most suitable model for the given dataset. The findings suggest that property size and modern amenities play a crucial role in determining house prices. These insights can help real estate businesses make informed decisions regarding property valuation, marketing strategies, and investment planning.